Object A is four times heavier than object B. Each object is attached to a spring, and both springs have the same spring constant. Both objects are then pulled from their equilibrium position and released from rest.

What is the ratio of the periods of the two oscillators if the amplitude of A is half the amplitude of B.

a) $T_A/T_B=2$ b) $T_A/T_B=1/4$ c) $T_A/T_B=4$ d) $T_A/T_B=1/2$

e) other (give the value)

A certain spring elongates 9.0 mm when it is suspended vertically and a block of mass *M* is hung on it. If driven by a force of variable frequency, at what frequency would resonance be obtained?

a) 0.088 rad/s b) 200 rad/s c) 33 rad/s d) 1140 rad/s
e) cannot be computed unless the value of *M* is given

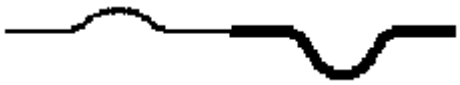
Two ropes are spliced together as shown and a pulse is traveling to the right in this rope.

A short time after the incident pulse shown in the diagram above, reaches the splice, what will be the rope's appearance?

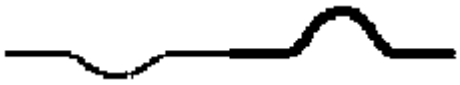
a)

b)

c)



d)



e)

A transverse wave traveling on a cord in the negative *x* direction has a wave number of 60 rad/m, a period of 0.20 s and an amplitude of 3.0 mm. What is the maximum transverse acceleration of a point on the cord

a) 180 mm/s^2 b) 60 mm/s^2 c) 4.87 m/s^2

d) 2.96 m/s^2 e) other (give the value)

A sinusoidal wave is traveling on a string with velocity 40 cm/s. The displacement of the particles of the string at x=10 cm is given by the equation y=(5.0 cm) sin[1.0π–(4.0π s^{-1}) t]. Determine the wavelength of the wave.

a) 20.0 cm b) 10.0 cm c) 3.14 cm

d) 0.314 m e) other (give the value)

A truck moving at 36 m/s passes a police car moving at 45 m/s in the opposite direction. If the frequency of the siren relative to the police car is 500 Hz, what is the frequency heard by an observer in the truck after the police car passes the truck? (The speed of sound in air is 343 m/s.)

a) 396 Hz

b) 636 Hz

c) 361 Hz

d) 393 Hz

e) 617 Hz

Question 7.

A wave generated in a medium is a longitudinal wave when

a) there is a net transport of matter by the wave.

b) the molecules of the medium are unable to exert forces on each other.

c) molecular displacements are parallel to the wave velocity.

d) molecular displacements are perpendicular to the wave velocity.

e) the density of the medium is less than the density of water.

A trumpet player hears 3 beats per second when she plays a note and simultaneously sounds a 440 Hz tuning fork. After pulling her tuning valve out to slightly increase the length of her trumpet, she hears 5 beats per second against the tuning fork. What was her initial frequency?

a) 437 Hz

b) 443 Hz

c) Either a) or b).

d) 435 Hz

e) 445 Hz

A vertical tube one meter long is open at the top. It is filled with 75 cm of water. If the velocity of sound is 344 m/s, what will the fundamental resonant frequency be (in Hz)?

a) 3.4

b) 172

c) 344

d) 1.7

e) 688

Two sinusoidal waves are described by $y_1 = 3\sin(94x - 700t)$ and $y_2 = 3\sin(94x + 700t)$ where the displacements y_1 and y_2 are measured in metres. What is the amplitude of the resultant wave when they interfere?

a) 8.0 m

b) 4.3 m

c) 3.0 m

d) 3.2 m

e) 6.0 m

A speaker produces a 440 Hz sound with total power 1.2 W, radiating equally in all directions. At a distance of 5.0 m determine,

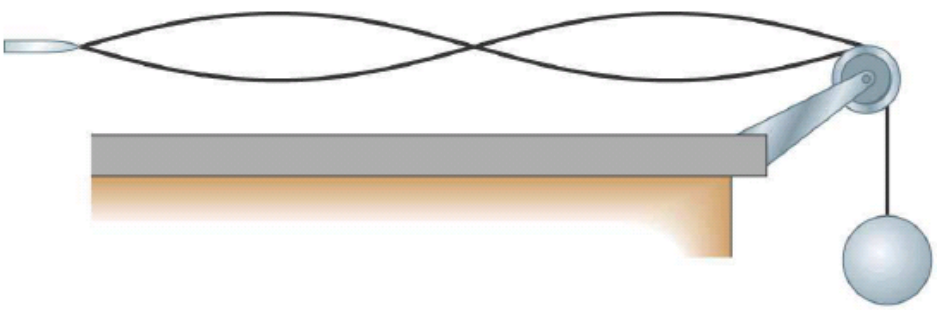
a) the average sound intensity

b) (the loudness

c) the pressure amplitude

d) the displacement amplitude.

One end of a horizontal string is attached to a vibrating blade with frequency 84 Hz and the other end passes over a pulley. A sphere of mass 2.00 kg hangs on the other end of the string. The mass per unit length of the string is 56 g/m. The string is vibrating in its second harmonic.



(a) Determine the length of the string on which the standing wave is formed.

(b) Assume that both the frequency of the vibrating blade and the length of the string remain the same. Determine the largest mass the sphere can have so that a standing wave can still be formed on the string.

(c) Knowing that the amplitude of the loops in 5 cm, write the mathematical equation for the standing wave with the two loops as shown in the figure above.